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Case assignment

Your task is to build an API for the fictious moving company MoveIT. You are free to choose any technology and framework, but please motivate your choice. The deliverable should be executable code.

We want to see code you have written, not code generated by the development environment.

Look upon this as an opportunity to show your skills!

Background

MoveIT has a long background of providing moving services, but lately they are experiencing that they lose potential customers because it’s complicated to get a a price offer. MoveIT have therefore decided to develop a web API, that clients can use for price inquiries.

This is the first step in MoveIT’s long term plan to be the brand of choice when moving. The company wants a solution that in the future could be sold as a service and they wish to see an architecture that could be easily extended.

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The task

The task is to develop the first version of a web API, that clients can consume in order to get price offers.

MoveIT has defined a number of business rules for price calculation.

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Business rules

Distance The price depends on the distance of the move, where the total distance makes the price calculation end up in one of the following categories:

- 1000 SEK + 10 SEK / km for distances above 50km
- 5000 SEK + 8 SEK / km for distances above 50km and below 100 km
- 10000 SEK + 7 SEK / km for distances above 100 km

Distance	Price
10 km	1100 SEK
49 km	1490 SEK
50 km	5400 SEK
51 km	5408 SEK
99 km	5792 SEK
100 km	10700 SEK

Table 1. Calculation of distance price

Volume

Given a volume, calculated from the number of square meters of the living area, one or more cars are needed. For every 50 m^2 an extra car is needed and each car costs the actual distance price calculated according to the table on the right. Basement and attic areas are expected to contain twice the volume as regular living areas.

Living area	Attic area	# of cars	Distance price / car	Price
49 m^2	0 m^2	1	1100 SEK	1100 SEK
10 m^2	25 m^2	2	1100 SEK	2200 SEK
50 m^2	0 m^2	2	1100 SEK	2200 SEK
100 m^2	0 m^2	3	1100 SEK	3300 SEK
150 m^2	0 m^2	4	1100 SEK	4400 SEK

Table 2: Calculation of volume price

Piano

Due to extra handling the price for moving a piano is an extra 5 000 SEK. This is regardless of distance and volume.

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Acceptance criterias

To solve the task you will need to ensure that the price is calculated according to the business rules.

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Instructions

Try to solve the task in 4-6 hours.

We focus on the code you have written and wish to see a clean solution. Remember the definition of clean code: Clean code is written by someone who cares.

Since there is a time limitation, we don’t expect perfection.

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Save the assignment

When done with the task, we wish you to upload the solution – including source code – to a shared folder on Valtech's Dropbox. Only you and people at Valtech directly involved will have access to the uploaded material.

Instructions on how to access the Dropbox folder will be provided to you.

- Use your personal link to open the shared folder
- Save all files included in your case directly to the folder or as a zip archive.
- Let us know when the files are uploaded

Don’t hesitate to get in touch with us if you have any questions.

Good luck!

Thank you!